

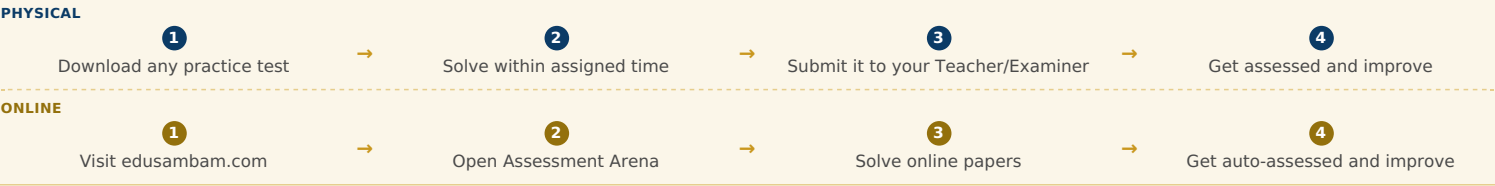
EDUSAMBAM ASSESSMENT SERIES

Physics Comprehensive Practice Test 1

PRINTABLE PRACTICE WORKSHEET

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YOUR PHYSICAL & ONLINE SELF-ASSESSMENT ROADMAP



Candidate Name:		Roll No.:	
Class / Grade:		Date:	
School / Institution:		Test Centre:	

Subject	Physics	Topic	Physics Comprehensive Practice Test 1	Total Marks	100
Total Questions	45	Time Allowed	1 Hour (60 minutes)	Version	A

GENERAL INSTRUCTIONS

1. Read each question carefully before answering.
2. For Section A and Section B, mark one answer only.
3. For Sections C and E, write your answer clearly in the space provided.
4. For Section D, write the correct letter from Column B against each item in Column A.
5. Do not use correction fluid. Keep all answers clear and legible.
6. This worksheet is prepared from the EDUSAMBAM practice assessment supplied for this paper.

PAPER PATTERN

Section	Questions	Marks Each	Total
Section A — Multiple Choice	15	2	30
Section B — True / False	10	2	20
Section C — Fill in the Blank	10	2	20
Section D — Matching	5 pairs	2	10
Section E — Short Answer	5	4	20
Total	45	—	100

Section A — Multiple Choice

30 marks

1. What is energy?

A. A type of force

B. The capacity to do work

C. Only heat and light combined

D. A unit of mass

Answer: ☐A ☐B ☐C ☐D

Basic • 2 pts

2. What is an electric current?

A. The resistance in a wire

B. A type of magnetic pole

C. The flow of electric charge through a conductor

D. A form of light

Answer: ☐A ☐B ☐C ☐D

Basic • 2 pts

3. What is a force?

A. A type of energy that never changes

B. A push or a pull on an object

C. The speed of an object

D. A measurement of mass only

Answer: ☐A ☐B ☐C ☐D

Basic • 2 pts

4. What is gravity?

A. A force that only exists on Earth

B. A force of attraction between any two objects with mass

C. A type of electromagnetic wave

D. A force that repels objects apart

Answer: ☐A ☐B ☐C ☐D

Basic • 2 pts

5. What does temperature actually measure?

A. The average kinetic energy of particles

B. The total amount of thermal energy in an object

C. The mass of a substance

D. The colour of an object

Answer: ☐A ☐B ☐C ☐D

Intermediate • 2 pts

6. What is light?

A. A liquid substance

C. A type of sound wave

Answer: ☐A ☐B ☐C ☐D

B. A form of electromagnetic energy that travels as a wave

D. Only visible during the day

7. What is matter?

A. Anything that is alive

C. Only solids and liquids

Answer: ☐A ☐B ☐C ☐D

B. Anything that has mass and takes up space

D. Only things you can see

8. What is the key difference between distance and displacement?

A. They are exactly the same thing

C. Distance includes direction; displacement does not

Answer: ☐A ☐B ☐C ☐D

B. Displacement includes direction; distance is only total path length

D. Displacement is always larger than distance

9. What is pressure?

A. The total weight of an object

C. The speed at which fluid moves

Answer: ☐A ☐B ☐C ☐D

B. Force spread over an area

D. A type of buoyant force

10. What is sound?

A. A form of light

C. A mechanical wave created by a vibrating object

Answer: ☐A ☐B ☐C ☐D

B. An electromagnetic wave

D. A type of force

11. What does the word "physics" originally come from, and what does it mean?

A. Greek "physika," meaning "natural things"

C. Arabic "fizik," meaning "number"

Answer: ☐A ☐B ☐C ☐D

B. Latin "physicus," meaning "doctor"

D. French "physique," meaning "body"

12. In physics, when is work actually being done on an object?

A. Whenever you feel tired

C. When a force causes an object to move in the direction of that force

Answer: ☐A ☐B ☐C ☐D

B. Whenever a force is applied, moving or not

D. Only when electricity is involved

13. What does the Law of Conservation of Energy state?

A. Energy cannot be created or destroyed, only transformed

C. Energy can be destroyed but never created

Answer: ☐A ☐B ☐C ☐D

B. Energy always increases over time

D. Only living things can create energy

14. What is the scattering of sunlight by gas molecules in the atmosphere called?

A. Refraction

C. Rayleigh scattering

Answer: ☐A ☐B ☐C ☐D

B. Reflection

D. Diffusion scattering

15. Doing 100 joules of work in 2 seconds gives how much power?

A. 200 watts

C. 100 watts

Answer: ☐A ☐B ☐C ☐D

B. 50 watts

D. 2 watts

Section B — True / False

20 marks

16. Energy cannot be created or destroyed, but it can be transformed from one form to another.

A. True ☐

B. False ☐

17. A solar panel transforms light energy into electrical energy.

A. True ☐

B. False ☐

18. In a series circuit, each component has a separate independent path for current.

A. True ☐

B. False ☐

19. Newton’s First Law is also called the law of inertia.

A. True ☐

B. False ☐

20. If the distance between two masses triples, their gravitational force becomes one-third as large.

A. True ☐

B. False ☐

21. Temperature and heat mean exactly the same thing in physics.

A. True ☐

B. False ☐

22. Refraction is the bending of light as it changes speed when entering a different medium.
A. True ☐ **B.** False ☐

23. Sound can travel through the vacuum of outer space.
A. True ☐ **B.** False ☐

24. An object can have zero displacement after travelling a non-zero distance.
A. True ☐ **B.** False ☐

25. A simple machine can reduce the total amount of work required by creating extra energy.
A. True ☐ **B.** False ☐

Intermediate · 2 pts

Intermediate · 2 pts

Intermediate · 2 pts

Intermediate · 2 pts

Section C — Fill in the Blank

20 marks

26. The SI unit of energy is the ____.
Answer: _____

27. Electric current is measured in ____.
Answer: _____

28. Newton’s Second Law is written as $F = m$ ____.
Answer: _____

29. The force that attracts masses toward one another is called ____.
Answer: _____

30. The three main methods of heat transfer are conduction, convection, and ____.
Answer: _____

31. The bending of light when it passes from one medium into another is called ____.
Answer: _____

32. The change from liquid to gas is called ____.
Answer: _____

33. Average speed is calculated as total distance divided by total ____.
Answer: _____

34. The upward force exerted by a fluid on an immersed object is called ____.
Answer: _____

35. Power is the rate at which ____ is done or energy is transferred.
Answer: _____

Basic · 2 pts

Basic · 2 pts

Basic · 2 pts

Basic · 2 pts

Intermediate · 2 pts

Intermediate · 2 pts

Intermediate · 2 pts

Intermediate · 2 pts

Intermediate · 2 pts

Intermediate · 2 pts

Section D — Matching

10 marks

36. Match each term to its correct definition.

Column A	Answer	Column B
Kinetic energy	_____	A. Bending of light between media
Inertia	_____	B. Rate of doing work
Refraction	_____	C. Energy due to motion
Buoyancy	_____	D. Upward force from a fluid
Power	_____	E. Resistance to a change in motion

Section E — Short Answer

20 marks

41. Explain the difference between kinetic energy and gravitational potential energy.

42. Why does a seatbelt protect a passenger when a car stops suddenly?

Advanced · 4 pts

Advanced · 4 pts

43. State the difference between mass and weight.

Advanced · 4 pts

44. Why does a steel ship float even though solid steel is denser than water?

Advanced · 4 pts

45. Explain why a ramp makes lifting a load easier without reducing the total work in the ideal case.

Advanced · 4 pts

Candidate Signature: _____ **Invigilator Signature:** _____

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Marks Obtained: _____ / 100

Checked By: _____

Result / Grade: _____

Date Checked: _____

EDUSAMBAM ASSESSMENT SERIES

Physics Comprehensive Practice Test 1 — Answer Key

Teacher reference: equivalent grammatically correct wording is acceptable for open-ended items when the meaning is preserved.

Section A — Multiple Choice

- | | |
|------|-------|
| 1. B | 9. B |
| 2. C | 10. C |
| 3. B | 11. A |
| 4. B | 12. C |
| 5. A | 13. A |
| 6. B | 14. C |
| 7. B | 15. B |
| 8. B | |

Section B — True / False

- | | |
|-----------|-----------|
| 16. False | 21. True |
| 17. False | 22. False |
| 18. True | 23. True |
| 19. False | 24. False |
| 20. True | 25. True |

Section C — Fill in the Blank

- | | |
|---------------|------------------|
| 26. joule | 31. refraction |
| 27. amperes | 32. vaporization |
| 28. a | 33. time |
| 29. gravity | 34. buoyancy |
| 30. radiation | 35. work |

Section D — Matching

- | | |
|------------------------|------------------|
| 36. Kinetic energy — C | 39. Buoyancy — D |
| 37. Inertia — E | 40. Power — B |
| 38. Refraction — A | |

Section E — Short Answer (Model Answers)

- 41.** kinetic; motion; potential; height
- 42.** inertia; force; slow; stop
- 43.** mass; weight; gravity
- 44.** hollow; displace; water; density; buoyant
- 45.** less force; longer distance; work; trade